

EE 3203: Power Electronics  
Tutorial 3: Single Phase Rectifiers

1. A half-wave SCR controlled rectifier with a 230 V, 50 Hz input is feeding a RL load with  $R = 5\ \Omega$  as shown in Fig. 1. The SCR is triggered at a firing angle of  $45^\circ$  and the load current extinguishes at an angle of  $200^\circ$ . Find the circuit turn off time  $t_c$ , average output voltage and the average load current.

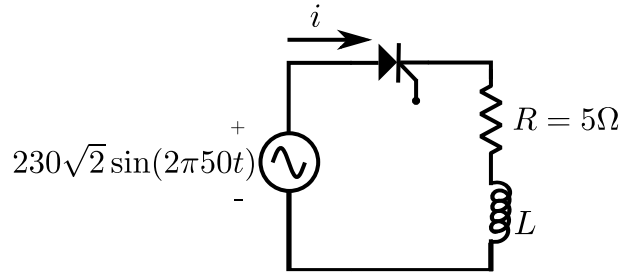


Fig. 1: Half-wave controlled rectifier feeding RL load.

2. Venu did a basic course in Power Electronics, got interested and built a single phase controlled full bridge rectifier with SCRs feeding a highly inductive load. The source inductance ( $L_s$ ) in his circuit was too low that it can be assumed to be zero. From his observation of the various circuit waveforms in an oscilloscope, he found that there was some hardware issue. He identified the issue to be arising from the gate drive of one of the four SCRs. Due to the hardware issue, one device alone was always being fired at an angle of  $0^\circ$  instead of the set  $\alpha$ , for all values of  $\alpha$ . Fig. 2 shows the controlled full bridge circuit with the hardware fault. The device with the faulty firing circuit is indicated as a diode  $D_3$  based on its operation.

Assuming a firing angle of  $\alpha = 30^\circ$ , sketch the following waveforms which Venu would have observed:  $v_s$ ,  $v_{D_3}$ ,  $i_{D_3}$ ,  $v_{T_4}$ ,  $i_{T_4}$ ,  $v_o$ ,  $i_s$ . Also, derive the expression for the output average voltage for a general firing angle  $\alpha$ .

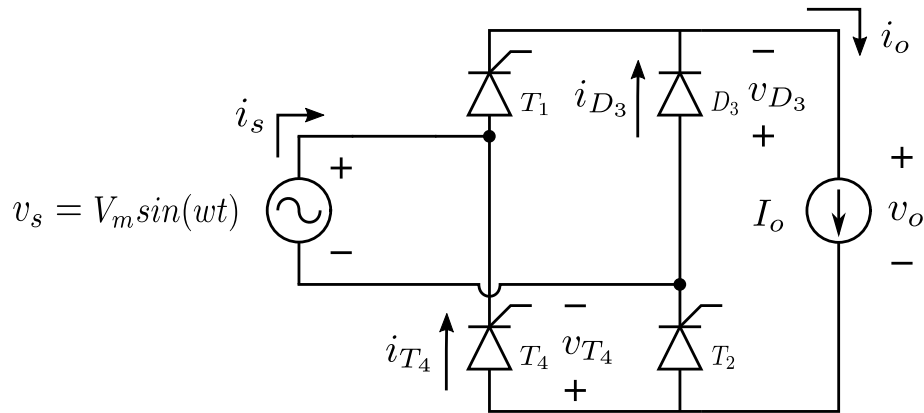


Fig. 2: Single-phase semi-controlled bridge rectifier.

3. Fig. 3 shows a series LC circuit with a semi-controlled switch. The initial current in the inductor and the voltage across the capacitor are zero and the SCR is off. At time  $t = 0^+$ , the SCR is turned on. Consider the switch to be ideal. The inductance  $L$  is  $20 \mu\text{H}$ ; the capacitance is  $20 \mu\text{F}$ .

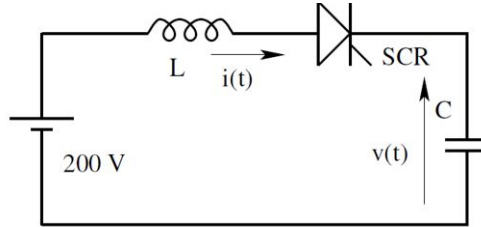


Fig. 3: A series LC circuit with a switch.

- Evaluate the inductor current  $i(t)$  and capacitor voltage  $v(t)$  following turn on.
  - Sketch  $i(t)$ , and  $v(t)$  for  $t > 0$  and mark the salient features.
  - Evaluate the final stored energy in the inductor and the capacitor.
  - Evaluate the peak stored energy in the inductor and the capacitor during the transient.
  - Evaluate the energy drawn from the source during the transient and carry out an energy balance of the system.
  - Evaluate the peak conducting current stress and peak blocking voltage stress on the SCR.
4. Fig. 4 shows the circuit of a battery charger. The power converter is a half controlled rectifier. The ac power input is at an ac voltage ( $V_{ac}$ ) of rms 110 V. The frequency of the ac input power is 50 Hz ( $f$ ). The current output of the rectifier is filtered by the output inductor  $L$ . The battery current is  $I(t)$ . The battery voltage is 72 V dc. Through appropriate control, the battery charging current is set for an average value ( $I_o$ ) of 10 A dc.

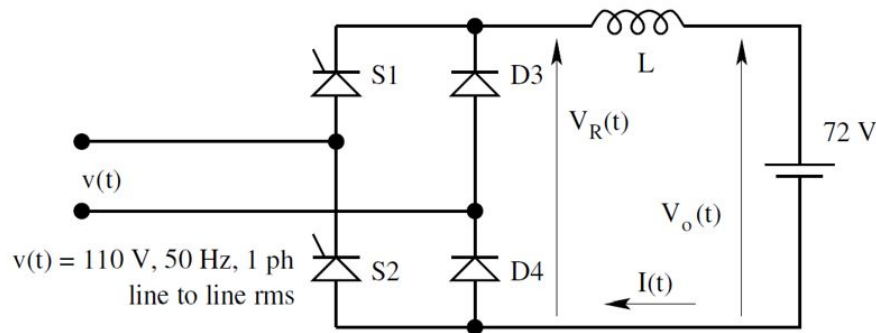


Fig. 4: Battery charger.

- a. Evaluate the control angle of the power converter under steady state operation of the battery charger.
- b. Sketch the rectifier output voltage  $V_R(t)$  and the battery charging current  $I(t)$  for one cycle under steady state and mark the salient points.
- c. Evaluate the ripple current  $I(t)$  as a function of L. Evaluate the value of L if the peak to peak ripple current in  $I(t)$  is to be limited to 10 % of  $I_o$ .